

BEYOND CHIPS

If start-ups are unlikely to build semiconductor fabs themselves, where will they create value? Investors believe the answer lies in the layers surrounding manufacturing. “Components and materials are the layer that India has almost entirely imported until now,” says Amit Chand, founder of BYT Capital. “The Electronics Component Manufacturing Scheme was designed specifically to close that gap, making it one of the least crowded opportunities for start-ups.”

Chand says the MPM Scheme is not a deep-tech story by itself. It primarily incentivises large-scale phone assembly but creates downstream demand for a supplier ecosystem. As mobile production scales up, demand for camera modules, batteries, connectors, testing equipment and other critical components—many of which India still imports—is expected to rise. “The opportunity is not on the assembly line itself but the domestic component and testing ecosystem that feeds it,” he says.

Another new category is deep-tech companies building the picks-and-shovels layer, such as inspection tools, specialty gases and materials, packaging IP, says Rajaram, adding that five years ago, a start-up building semiconductor manufacturing equipment or advanced materials in India had almost no domestic anchor customer. On had to sell in Europe or the US, which is a brutal ask for an early-stage company. Now, there’s a growing domestic market to sell into, along with capital support to de-risk the R&D.

THE NEXT PHASE

The policy shift extends beyond incentives. Earlier this year, the government revised the recogni-



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ASHOK CHANDAK

PRESIDENT, IESA AND SEMI INDIA



“Manufacturing is never going to be a start-up game. Start-ups can create value in chip design, intellectual property and specialised products”

SHASHWATH T. R.

CO-FOUNDER AND CEO, MINDGROVE TECHNOLOGIES

tion framework for deep-tech start-ups, extending the eligibility window from 10 years to 20 years and raising the turnover threshold from ₹200 crore to ₹300 crore. The move reflects an acknowledgement that deep-tech companies require longer gestation periods and take more time to commercialise than software start-ups.

However, the optimism comes with caveats. Building semiconductor capacity takes. While policy support and venture capital are improving, founders still face long development cycles, dependence on imported equipment and materials, and the challenge of commercialising complex technologies.

Shashwath cautions against expecting overnight results. “Building semiconductor products is a long cycle. Even with policy support, start-ups need access to manufacturing, testing, supply chains and customers before they can scale. Those parts of the ecosystem are still evolving.”

Whether India’s semiconductor

push ultimately produces the next NVIDIA or ASML is a question that will take years to answer. But industry leaders agree that the immediate objective is not to create a handful of successful companies. It is to create the ecosystem that allows hundreds of them to emerge.

Chandak believes the real success of the policy will lie in how each layer of the value chain begins to reinforce the next. “Components feed electronics manufacturing and semiconductors feed electronics manufacturing. As electronics manufacturing and exports grow, they create demand for components and semiconductors, encouraging more investment across the ecosystem,” he says. “That is how the virtuous cycle begins.”

Much like UPI created the digital infrastructure that enabled India’s fintech boom, Semicon India 2.0 and MPM Scheme are attempting to build the industrial infrastructure for deep tech. **BT**

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